

A Two-Transaction Counterexample to Repeated One-Step Oracle Sampling in a Risk-Limiting Financial Audit

Richie Sater

August 11, 2026

Abstract

Shekhar, Xu, Lipton, Liang, and Ramdas derive a one-step oracle rule for risk-limiting financial audits: sample a remaining transaction i with probability proportional to its monetary misstatement contribution $\pi_i f_i$. They do not claim that repeating this rule minimizes total expected audit length. We show that this derived global-optimality conjecture is false for their released ApproxKelly initialization after intersection with the logical confidence sequence. An exact two-transaction instance has expected stopping time $3/2$ under the repeated oracle, whereas the implementable proportional-to-recorded-value rule has expected stopping time $5/4$. The calculation uses rational arithmetic, enumerates every terminal history, and extends to an infinite rational family.

1 Setup and the conjecture

Consider N transactions with normalized recorded-value weights $\pi_i > 0$, where $\sum_i \pi_i = 1$, and fixed unknown misstatement fractions $f_i \in [0, 1]$. The total monetary misstatement is

$$m^* = \sum_{i=1}^N \pi_i f_i.$$

At time t , an auditor samples one remaining transaction from a predictable distribution q_t , observes its f_i , updates a confidence sequence, and stops when the confidence-set diameter is at most ε .

Shekhar et al. [1] prove that, conditional on the current history, the importance-weighted next observation has minimum variance under

$$q_t^{\text{oracle}}(i) = \frac{\pi_i f_i}{\sum_{j \in R_t} \pi_j f_j}.$$

This also maximizes a lower bound on one-step test-wealth growth. They describe the full fastest-uncertainty-reduction problem as a difficult multistage optimization and solve the one-step surrogate instead. The natural derived conjecture is that repeating the displayed oracle rule minimizes $\mathbb{E}[\tau_\varepsilon]$ globally. This conjecture, not a theorem or explicit conjecture of [1], is the target below.

2 The pinned confidence construction

For a candidate total m , the paper's betting process has the update

$$W_t(m) = W_{t-1}(m) \{1 + \lambda_t(m)(Z_t - \mu_t(m))\}, \quad W_0(m) = 1,$$

and the betting confidence set retains candidates with $W_t(m) < 1/\delta$. The logical interval after history $I_1, \dots, I_t$ is

$$[L_t, U_t] = \left[\sum_{s=1}^t \pi_{I_s} f_{I_s}, \sum_{s=1}^t \pi_{I_s} f_{I_s} + \sum_{i \in R_{t+1}} \pi_i \right].$$

We pin the construction to the authors' released implementation [2]: its ApproxKelly previous-payoff accumulator is initialized at zero, its first bet is therefore $\lambda_1 = 0$, and the betting bounds are intersected with the logical interval and their running intersection. This is also the logical-CS refinement in equation (6) of [1].

Lemma 1 (First-round reduction). *For the pinned construction and every $\delta \in (0, 1)$, the first-round betting confidence set is $[0, 1]$. Consequently, after sampling $I_1 = i$, the combined confidence-set diameter is $\sum_{j \neq i} \pi_j$.*

Proof. Since $\lambda_1 = 0$, one has $W_1(m) = 1$ for every $m \in [0, 1]$. The inequality $\delta < 1$ gives $1 < 1/\delta$, so no candidate is rejected. Intersecting $[0, 1]$ with $[L_1, U_1]$ leaves the logical interval, whose diameter is the sum of the remaining weights. $\square$

3 An exact counterexample

Theorem 1. *For the pinned construction, the repeated oracle policy does not minimize expected audit length over randomized without-replacement policies.*

Proof. Take $N = 2$, $\varepsilon = 1/3$, $\delta = 1/20$, and the following exact population:

i	π_i	f_i	$\pi_i f_i$
1	3/4	1/3	1/4
2	1/4	1	1/4

The oracle distribution at time one is therefore $(1/2, 1/2)$. Compare it with the prop-M rule of [1], which samples proportionally to π_i and hence uses $(3/4, 1/4)$.

If item 1 is sampled first, the first-round diameter is $1/4 \leq 1/3$, so the audit stops at time one. If item 2 is sampled first, the diameter is $3/4 > 1/3$, so it does not stop. After the other item is sampled at time two, the logical interval is a singleton and the audit stops. Thus a policy that samples item 1 first with probability q has

$$\mathbb{E}[\tau] = q + 2(1 - q) = 2 - q.$$

It follows that

$$\mathbb{E}[\tau(q^{\text{oracle}})] = 2 - \frac{1}{2} = \frac{3}{2}, \quad \mathbb{E}[\tau(q^{\text{prop-M}})] = 2 - \frac{3}{4} = \frac{5}{4}.$$

Since $5/4 < 3/2$, prop-M is strictly better by $1/4$ expected transaction. $\square$

The mechanism is transparent. The oracle makes the importance-weighted monetary contribution constant, so it has zero one-step variance. The logical diameter, however, is controlled by remaining recorded-value weight. On this instance, variance minimization is indifferent between the two items while logical-width reduction strongly favors item 1.

4 An infinite family

Proposition 1. *Let $p \in \mathbb{Q} \cap (1/2, 1)$, and set*

$$(\pi_1, \pi_2) = (p, 1 - p), \quad (f_1, f_2) = \left(\frac{1 - p}{p}, 1 \right).$$

Choose rational ε with $1 - p \leq \varepsilon < p$, and choose any rational $\delta \in (0, 1)$. Then

$$\mathbb{E}[\tau(q^{\text{oracle}})] = \frac{3}{2}, \quad \mathbb{E}[\tau(q^{\text{prop-M}})] = 2 - p,$$

so prop-M improves the expectation by $p - 1/2 > 0$.

Proof. Both contributions equal $1 - p$, making the oracle uniform. The condition on ε says that sampling item 1 first stops the audit and sampling item 2 first does not. The first-round probabilities are $1/2$ under the oracle and p under prop-M. Substitution in $\mathbb{E}[\tau] = 2 - q$ proves the claim. $\square$

5 Scope and verification

The theorem is deliberately narrow about the inferential construction. It does not address a betting CS without logical intersection, a different first-round betting initialization, or the imperfect-information minimax problem driven by estimated risk scores. Nor does it assert that prop-M is globally optimal. It suffices to refute the universal claim for the released construction.

The accompanying repository [3] represents every input and path probability as a reduced rational number. Its dynamic program enumerates the two terminal histories for each policy and regenerates a JSON certificate. No simulation, floating-point comparison, or numerical confidence-set grid is used in the proof.

References

- [1] S. Shekhar, Z. Xu, Z. Lipton, P. Liang, and A. Ramdas, “Risk-limiting financial audits via weighted sampling without replacement,” in *Proceedings of the 39th Conference on Uncertainty in Artificial Intelligence*, PMLR 216 (2023), 1932–1941.
<https://proceedings.mlr.press/v216/shekhar23a.html>.
- [2] S. Shekhar and Z. Xu, *WeightedWoRConfSeq*, commit a834e459 (2023).
<https://github.com/sshekhar17/WeightedWoRConfSeq/tree/a834e459a47f9efa74fa1706f1b1dd8173ffa30b>.
- [3] R. Sater, *RLFA Optimal Policy: an Exact Counterexample* (2026).
<https://github.com/RichieSater/rlfa-optimal-policy>.